Transforming Field Operations through AI-Powered Equipment and Workforce Management



OVERVIEW



Revolutionizing manual field operations into a streamlined digital process, we developed an integrated mobile app and web portal solution. The mobile app empowers remote equipment operators with intelligent tools for time tracking, equipment usage, and maintenance. The web portal enables managers to monitor activities, assign resources, manage HRMS, control inventory, and ensure maintenance efficiency.

ABOUT CLIENT



Our client is a leading construction equipment rental company offering a diverse fleet of heavy machinery including earthmoving machines, bridge cranes, piling rigs, Dozers, excavators and more. They serve multiple industries such as construction, mining, logistics, and energy, helping clients to enhance productivity.

CHALLENGES



Inefficient Manual Tracking ➤ Time logs and machine usage readings were inaccurate and hard to verify.

Lack of Real-Time Visibility ➤ No control over field activities and machine usage.

Coordination Gaps ➤ Paper-based tracking delayed field-equipment coordination.

Safety Compliance Issues ➤ Difficulty ensuring operators adhered to safety protocols on remote sites.

Poor Inventory Oversight ➤ Poor tracking of parts delayed maintenance.

SOLUTION



Mobile App Modules (for Field Operators & Mechanics)

- Operators can use app to clock in and out within geo-fenced zones, record timesheets, manage work orders, capture AI-verified machine logs, and submit reports with signatures.
- Mechanics manage service requests and maintenance via the app, which also supports attendance, expense claims, employee rosters, and real-time location tracking.

Web Portal Modules (for Managers & Admins)

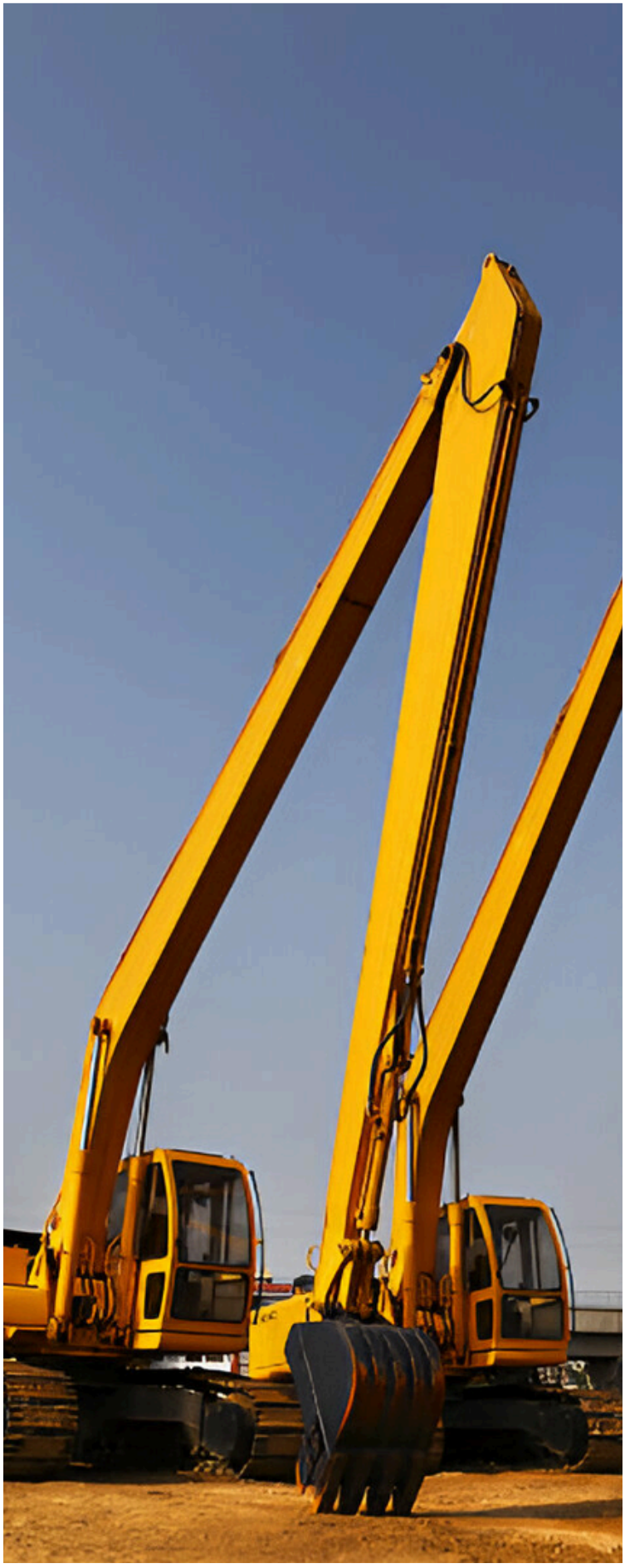
- The web portal handles HR onboarding, attendance, timesheet approvals, machine and employee scheduling, vehicle tracking, and inventory management with full control over material and stock flow.
- The web solution provides real-time analytics on equipment use, maintenance, and performance, along with incident and safety management to track and reduce onsite risks.



KEY FEATURES



- Smart geofencing for automated, accurate attendance.
- AI-enabled capture of machine readings with perfect accuracy.
- Real-time insights on usage, maintenance, and performance.
- Real-time inventory and procurement workflows.
- Verified digital log sheets with image/signature.
- Centralized tracking of equipment and maintenance.



RESULTS & IMPACTS

- 95% Reduction in manual errors across HRMS and operations.
- 90% Boost in attendance accuracy.
- 80% Full visibility & control over remote site activities.
- 70% Rise in operational coordination.

CONCLUSION

Our AI-powered mobile and web-based digital solution replaced the client's outdated manual methods with a unified operational ecosystem. From geofenced attendance, AI-verified machine usage readings, and mobile maintenance logging to centralized HRMS, inventory, and procurement workflows, the platform brought accuracy, speed, and visibility to every layer of the business.

